

Figure 1: **Late Crystallization:** Correct answers remain invisible in logit space throughout intermediate layers, then abruptly “crystallize” into top predictions at the final layer.

MECHLENS: Late Crystallization of Factual Knowledge Explains Intervention Effectiveness in Language Models

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Abstract

Understanding where LLMs store factual knowledge is critical for hallucination mitigation. We discover **Late Crystallization**: factual knowledge does not gradually emerge across layers but “crystallizes” abruptly at the final layers. In Qwen2.5-7B, 85.9% of correct answers *never* enter top-10 predictions at any intermediate layer. Cross-architecture validation (Llama-3.1-8B: 71.0%, Mistral-7B: 27.1%) confirms late emergence (>80% depth) is universal, while strict final-layer crystallization varies with attention mechanism. Cross-scale (Qwen2.5-14B: 95.8% FEP Depth) and cross-benchmark (MMLU: 98.2%) validation confirm generality; tuned lens comparison rules out probe artifacts ($\Delta=0.2\%$).

Late Crystallization explains an **architecture-dependent intervention pattern**: CAA outperforms DOLA on moderate-crystallization models (Llama: +33.5% vs. +1.9%), while DOLA dominates on high-crystallization Qwen (+25.4% MC1). LayerNorm ablation shows crystallization is intrinsic to the residual stream; LN scaling ($\times 1.2$) yields +11.8% MC1 with zero overhead. We also reveal a **Computability–Memorization Spectrum**: computable knowledge crystallizes earlier (layer 22.1/28) than memorized facts (28.0/28). We release MECHLENS supporting five model families.¹

1 Introduction

Large language models (LLMs) frequently produce plausible-sounding but factually incorrect outputs—hallucinations—posing significant challenges for deployment in high-stakes applications (Huang et al., 2023; Ji et al., 2023). A natural approach to mitigating hallucinations is to identify the internal components responsible for factual recall and intervene directly on their activations. However, the effectiveness of such interventions depends critically on whether factual knowledge is *localized* in specific model components or *distributed* across the network.

Prior work on knowledge localization has produced mixed evidence. Causal tracing (Meng et al., 2022) identifies middle-layer MLPs as critical for factual recall in GPT-2, motivating localized editing methods like ROME and MEMIT. However, recent work suggests knowledge may be more distributed than initially assumed (Geva et al., 2022), raising questions about whether activation-level interventions can effectively modulate factual behavior.

¹Code: <https://anonymous.4open.science/r/MechLens-COLM2026>

	Prior Work	This Work
Observation	Qualitative	Quantified (85.9%)
Metric	None	FEP (formal)
Cross-arch	Limited	3 architectures
Causal test	None	LN ablation
Intervention link	None	Predictive theory

Table 1: Comparison with prior logit lens studies.

We introduce MECHLENS, a unified mechanistic interpretability framework that enables systematic investigation of these questions. Through comprehensive experiments combining logit lens analysis, layer contrasting, and activation interventions, we discover a phenomenon we term **Late Crystallization**: factual knowledge does not gradually emerge in intermediate layers but crystallizes abruptly at the final layers (Figure 1). Our contributions:

- **Late Crystallization**: 85.9% of correct answers never enter top-10 at any intermediate layer, validated across architectures, scales (7–14B), benchmarks (TruthfulQA, MMLU), and probing methods (logit lens vs. tuned lens, $\Delta=0.2\%$)
- **Architecture-dependent intervention pattern**: CAA outperforms DOLA on moderate-crystallization models ($p<0.001$); DOLA dominates on high-crystallization Qwen (+25.4% MC1)
- **Causal evidence**: LayerNorm ablation shows crystallization is intrinsic to the residual stream; LN scaling ($\times 1.2$) yields +11.8% MC1 with zero overhead
- **Computability–Memorization Spectrum**: Computable knowledge crystallizes at layer 22.1/28; memorized facts only at 28.0/28

Together, these results bridge language model *understanding* (when does factual knowledge become explicit?), *theory* (why do different interventions succeed or fail?), and *practice* (how should practitioners select interventions based on architecture?).

2 Related Work

Knowledge Localization and Editing. Causal tracing (Meng et al., 2022) identifies middle-layer MLPs as critical for factual recall in GPT-2, motivating ROME and MEMIT (Meng et al., 2023). Geva et al. (2021; 2022) show MLPs function as key-value memories. Circuit tracing (Anthropic, 2025) reveals distributed factual circuits, while logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023) enable layer-wise prediction tracking. Concurrent work (Wang, 2025) extends logit lens tooling to modern architectures and observes late-layer knowledge concentration. Table 1 summarizes how our work extends beyond prior observations.

Activation Intervention for Factual Accuracy. ITI (Li et al., 2023) shifts activations along probed truthfulness directions. DOLA (Chuang et al., 2024) contrasts early/late logit distributions without learned directions. CAA (Turner et al., 2023) and representation engineering (Zou et al., 2023) steer via contrastive activation vectors. SADI (Wang et al., 2025) achieves state-of-the-art MC1=67% via semantic-adaptive intervention on instruction-tuned models. CCS (Burns et al., 2023) discovers latent knowledge through consistency constraints. These methods operate in activation space (ITI, CAA, SADI) or logit space (DOLA); our work provides a mechanistic account of *why* their effectiveness differs across architectures.

3 The MECHLENS Framework

MECHLENS is built on TransformerLens (Nanda & Bloom, 2022) and supports five model families with diverse architectures (Table 2).

Model	Layers	Heads	d	Attention
Qwen2.5-0.5B	24	14	896	GQA (2 KV)
Qwen2.5-7B	28	28	3584	GQA (4 KV)
Qwen2.5-14B	48	40	5120	GQA (8 KV)
Pythia-1.4B	24	16	2048	MHA
Llama-3.1-8B	32	32	4096	GQA (8 KV)
Mistral-7B	32	32	4096	GQA+SWA (8 KV)

Table 2: Supported model architectures. Qwen2.5-14B enables cross-scale validation (§7.2).

Analysis. We extend causal tracing (Meng et al., 2022) with adaptive noise calibration ($\sigma = \alpha \cdot \text{std}(\mathbf{E}(x))$, $\alpha=10$), KL divergence metrics, and multi-run averaging ($n=5$). We also implement contrastive activation analysis, computing normalized L2 distance between correct and hallucinated residual stream activations at each layer. Full methodological details are in Appendix B.

Intervention. MECHLENS supports three reversible intervention types through hooks: ablation ($\mathbf{h}'_{l,c} = \mathbf{0}$), scaling ($\mathbf{h}'_{l,c} = \alpha \cdot \mathbf{h}_{l,c}$), and injection ($\mathbf{h}'_{l,c} = \mathbf{h}_{l,c}^{\text{source}}$). For CAA, we follow Turner et al. (2023): given paired truthful/untruthful prompts, we extract contrastive directions $\mathbf{d}_l = \frac{1}{N} \sum_i (\mathbf{h}_l(x_i^+) - \mathbf{h}_l(x_i^-))$ and add $\alpha \cdot \mathbf{d}_l$ to the top- k layers ranked by $\|\mathbf{d}_l\|_2$ at inference time. Directions are extracted from 256 TruthfulQA training pairs.

4 Experimental Setup

Models and Phases. *Phase 1* (causal tracing, contrastive analysis, activation scaling) uses Qwen2.5-0.5B and Pythia-1.4B—matched 24-layer models with different architectures (GQA/SwiGLU vs. MHA/GELU). *Phase 2* (FEP detection, MC1/MC2 evaluation, cross-architecture validation) uses Qwen2.5-7B, Llama-3.1-8B, and Mistral-7B on the full TruthfulQA benchmark (817 samples) (Lin et al., 2022). *Phase 3* (cross-scale validation) extends FEP detection to Qwen2.5-14B (48 layers) to test whether Late Crystallization persists at larger scale.

Evaluation. Phase 1 screening uses 200 TruthfulQA samples with keyword matching plus LLM-as-Judge (Zheng et al., 2023). Phase 2 uses standardized MC1 (single-correct accuracy) and MC2 (normalized probability mass) on all 817 samples. We test 20 activation scaling strategies across 6 categories plus an 18-configuration ITI grid search; full details and results are in Appendix A.

5 Results

5.1 Negative Results: Activation Scaling and ITI

Simple activation scaling universally fails: only 2 of 20 strategies show any positive effect ($\leq 2\%$), while aggressive scaling increases hallucination by up to 28% (Qwen2.5-7B). An 18-configuration ITI grid search achieves at most +2% (Pythia-1.4B) and 0% (Qwen). This consistency across 4 models, 2 benchmarks, and 38 configurations demands a mechanistic explanation (Section 6; full tables in Appendix A).

5.2 MC1/MC2 Evaluation with Advanced Interventions

We evaluate four methods on the full 817-sample TruthfulQA benchmark using Qwen2.5-7B (Table 3).

Key findings. A clear hierarchy emerges: DoLA dynamic (+25.4% MC1) > CAA top_k=10 (+15.5%) > ITI top_k=10 (+10.0%) > simple scaling (0%). Both ITI and CAA show zero

Method	Configuration	MC1	MC2
Baseline	—	0.2215	0.3921
ITI	top_k=3, $\lambda=1-3$	0.2215	0.3921
ITI	top_k=5, $\lambda=1-3$	0.2203	0.3926
ITI	top_k=10, $\lambda=3$	0.2436	0.4178
DoLA	early (static)	0.2326	0.4371
DoLA	mid (static)	0.2448	0.4555
DoLA	dynamic	0.2778	0.4822
CAA	top_k=3, all coeff	0.2215	0.3921
CAA	top_k=5, all coeff	0.2203	0.3922
CAA	top_k=10, coeff=5.0	0.2558	0.4338

Table 3: MC1/MC2 on TruthfulQA (Qwen2.5-7B, 817 samples). DoLA dynamic achieves the best MC1 (+25.4% over baseline, $p=0.009$). 95% bootstrap CI for MC1 $\approx \pm 0.03$ ($n=817$).

improvement at top_k $\in \{3, 5\}$ but substantial gains at top_k=10, indicating a layer-count threshold. That DoLA (logit-space) achieves the largest gains motivates our mechanistic analysis in Section 6.

Base model baselines. All experiments use base (non-instruction-tuned) models (MC1 baselines: 18.9–22.2%). Absolute values should not be compared with instruction-tuned methods like SADI (Wang et al., 2025) (MC1=67%); our focus is on the *relative* effectiveness hierarchy. Two-proportions z-test confirms DoLA’s improvement is significant ($z=2.63$, $p=0.009$). With Bonferroni correction for 3 pairwise comparisons (adjusted $\alpha=0.017$), DoLA vs. baseline remains significant ($p=0.009 < 0.017$).

6 Late Crystallization of Factual Knowledge

The effectiveness hierarchy observed in Section 5.2—and the striking failure of simple activation scaling—demands a mechanistic explanation. We hypothesize and test the idea that factual knowledge in LLMs undergoes a phase transition at the final layers, which we term **Late Crystallization**.

6.1 Factual Emergence Point Detection

We define the **Factual Emergence Point (FEP)** for a query q with correct answer a as the earliest layer L at which a enters the top- k predictions under logit lens projection:

$$L_{\text{FEP}} = \min\{L : \text{rank}(a, \mathbf{W}_U \cdot \text{LN}(\mathbf{h}_L)) \leq k\} \quad (1)$$

where $\mathbf{h}_L$ is the residual stream at layer L , LN is the final layer norm, and $\mathbf{W}_U$ is the unembedding matrix. If a never enters top- k at any intermediate layer, we set $L_{\text{FEP}} = L_{\text{max}}$ (the final layer). We use $k = 10$ throughout.

Tuned lens validation. We use the standard logit lens (nostalgebraist, 2020) rather than the tuned lens (Belrose et al., 2023) because it requires no additional training. A natural concern is that FEP = L_{max} for 85.9% of samples could reflect logit lens’s known limitations as a probe rather than a genuine phenomenon. Four lines of evidence—including direct empirical comparison—argue against this:

(1) *Direct tuned lens comparison:* We train per-layer affine probes $\mathbf{h}'_l = \mathbf{h}_l \mathbf{W}_l + \mathbf{b}_l$ (identity-initialized) on 2,000 WikiText-2 samples using KL divergence against final-layer logits (Adam optimizer, lr= 10^{-3} , batch size 32, 3 epochs), then compute tuned lens FEP on all 817 TruthfulQA samples (Table 4). The tuned lens yields a late crystallization rate of 85.7%, within 0.2 percentage points of the logit lens (85.9%). Of 817 samples, 74.9% have identical FEP under both methods. The tuned lens does detect earlier emergence (mean FEP=25.1 vs. 27.3, $\sigma=7.4$ vs. 1.8), primarily for high-frequency answer tokens (e.g., “The”) where learned

Probe	Mean FEP	Late Crystal	Exact Match	± 2 Match
Logit lens	27.3 $\pm$ 1.8	85.9%	74.9%	76.4%
Tuned lens	25.1 $\pm$ 7.4	85.7%		

Table 4: Tuned lens vs. logit lens FEP (Qwen2.5-7B, 817 samples). Both methods yield near-identical late crystallization rates ($\Delta=0.2\%$), confirming the phenomenon is not a probe artifact.

k	Mean FEP	σ	Late Crystal	FEP Depth
1	28.0	0.0	100.0%	100.0%
3	27.9	0.4	98.3%	99.8%
5	27.6	1.4	92.9%	98.7%
10	27.3	1.8	85.9%	97.5%
20	27.0	2.1	80.2%	96.5%
50	26.6	2.6	72.7%	95.0%
100	26.0	3.4	64.7%	93.0%

Table 5: FEP sensitivity to k (Qwen2.5-7B, 817 samples). Late Crystallization is robust: even at $k=100$, 64.7% of answers never appear at intermediate layers. FEP Depth stays $>93\%$ across all thresholds.

137 affine corrections amplify weak early-layer signal—consistent with the probes functioning
138 correctly.

139 (2) *LayerNorm ablation* (Section 7.5): Ablating `ln_final` produces identical FEP distributions
140 (Mean FEP=27.30, 85.9% late crystallization), ruling out LN projection mismatch as a con-
141 found.

142 (3) *Cross-architecture systematicity*: If Late Crystallization were merely a probe artifact, we
143 would expect random variation across architectures. Instead, we observe a systematic
144 gradient—Qwen (85.9%) $>$ Llama (71.0%) $>$ Mistral (27.1%)—that correlates with attention
145 mechanisms (standard GQA vs. sliding window). Sliding window attention enables earlier
146 information routing, producing measurably lower crystallization, which an artifact theory
147 cannot explain.

148 (4) *Intervention–FEP correlation*: The crystallization degree *predicts* which intervention
149 dominates (DOLA on high-crystallization Qwen, CAA on moderate-crystallization
150 Llama/Mistral; Section 7). A poor probe would yield FEP values uncorrelated with down-
151 stream intervention success.

152 We compute FEP for all 817 TruthfulQA samples on Qwen2.5-7B (28 layers), tracking the
153 rank of each correct answer’s first token across all layers via logit lens projection.

154 **Robustness to k threshold.** A natural concern is whether our findings depend on the
155 specific choice of $k=10$. Table 5 shows FEP statistics across $k \in \{1, 3, 5, 10, 20, 50, 100\}$. Late
156 Crystallization is *robust*: even at $k=100$ (top 0.1% of vocabulary), 64.7% of correct answers
157 still never appear at any intermediate layer, and FEP Depth remains above 93% for all
158 thresholds. The monotonic relationship between k and late crystallization rate confirms a
159 genuine phenomenon rather than a threshold artifact.

160 6.2 The Late Crystallization Phenomenon

161 Table 6 reveals a striking pattern: **85.9% of correct answers (702/817) never enter the top-10**
162 **predictions at any intermediate layer.** The mean FEP is 27.3 ± 1.8 out of 28 layers. Only
163 14.1% show any intermediate-layer emergence, with layer 23 the most common non-final
164 point (5.8%).

165 This **Late Crystallization** reveals that factual knowledge does not gradually “build up”
166 across depth. Correct answers exist in a distributed, implicit form within the residual stream

FEP Layer	Count	Percentage
28 (final / never)	702	85.9%
23	47	5.8%
21	20	2.4%
25	13	1.6%
22	10	1.2%
24	10	1.2%
26	10	1.2%
20	5	0.6%

Table 6: FEP distribution (Qwen2.5-7B, 817 TruthfulQA samples). 85.9% of correct answers *never* enter top-10 at any intermediate layer. Mean FEP = 27.3 ± 1.8 .

(invisible to logit lens) and abruptly “crystallize” into explicit predictions only at the final layer.

6.3 Computability–Memorization Spectrum

FEP varies systematically across knowledge categories (full table in Appendix F): Logical Falsehood crystallizes earliest (mean FEP=22.1, $\sigma=2.6$), while categories like History, Psychology, and Weather show FEP=28.0 with *zero* variance—every sample has its answer invisible until the final layer. This **Computability–Memorization Spectrum** reveals that “computable” knowledge emerges at intermediate layers, while purely memorized world knowledge remains fully distributed until crystallization.

6.4 FEP–DoLA Decorrelation

DoLA’s dynamic layer selection does not correlate with FEP (Pearson $r = -0.057$, $p = 0.103$). DoLA selects near-exclusively layer 0–1 (mean = 0.95), while FEP concentrates at layer 27–28. This reveals DoLA’s true mechanism: contrasting the maximally early layer (surface patterns) against the final layer (crystallized knowledge), amplifying the factual signal that crystallization produces.

Late Crystallization explains all observed patterns: DoLA contrasts pre- vs. post-crystallization logits; CAA/ITI require $\text{top}_k \geq 10$ to span the pre-crystallization window; simple scaling amplifies all components uniformly, unable to selectively steer the nonlinear crystallization process. The $\text{top}_k=10$ threshold corresponds to $\sim 36\%$ of Qwen2.5-7B’s 28 layers, spanning the final third where implicit knowledge is distributed (layers 18–28). With $\text{top}_k \leq 5$, interventions affect only the post-crystallization region where answers are already visible; with $\text{top}_k=10$, interventions capture the implicit factual signal distributed across the pre-crystallization residual stream.

7 Cross-Architecture Validation and Causal Analysis

We extend our analysis to Llama-3.1-8B (32 layers, GQA) and Mistral-7B (32 layers, GQA+SWA).

7.1 Cross-Architecture FEP Detection

Table 7 and Figure 2 reveal two key findings: (1) **Late knowledge emergence is universal**—across all four models (including 14B), factual knowledge emerges at $>80\%$ model depth; (2) **The degree varies with architecture**—strict final-layer crystallization ranges from 85.9% (Qwen-7B) to 27.1% (Mistral), where sliding window attention appears to enable earlier information routing.

Model	Layers	Mean FEP	FEP Depth	Late Crystal
Qwen2.5-7B	28	27.3 ± 1.8	97.5%	85.9%
Qwen2.5-14B	48	46.0 ± 4.9	95.8%	77.7%
Llama-3.1-8B	32	29.4 ± 4.9	91.9%	71.0%
Mistral-7B	32	26.3 ± 6.2	82.3%	27.1%

Table 7: Cross-architecture and cross-scale FEP detection (817 samples). FEP Depth = Mean FEP / total layers. All models show knowledge emergence >80% depth.

Benchmark	n	Mean FEP	Late Crystal
TruthfulQA	817	27.3 ± 1.8	85.9%
MMLU (all)	1,200	27.9 ± 0.5	98.2%
STEM	450	28.0 ± 0.4	99.1%
Humanities	250	28.0 ± 0.4	98.0%
Social Sciences	250	27.9 ± 0.7	97.2%
Other	250	27.9 ± 0.6	98.0%

Table 8: Cross-benchmark FEP validation (Qwen2.5-7B). Late Crystallization is even more pronounced on MMLU (98.2%) than TruthfulQA (85.9%), and consistent across all MMLU subject groups (97.2–99.1%).

7.2 Cross-Scale Validation (Qwen2.5-14B)

To test whether Late Crystallization persists at larger scale, we run FEP detection on Qwen2.5-14B (48 layers, 14.7B parameters)—the same architecture family as Qwen2.5-7B but with 71% more layers and $2\times$ the parameters. Results (Table 7, row 2) show that Late Crystallization *persists*: 77.7% of correct answers never enter top-10 at any intermediate layer, and FEP Depth is 95.8% (vs. 97.5% at 7B). The 1.7 percentage point difference in FEP Depth is negligible, confirming that the relative layer at which knowledge crystallizes is *scale-invariant* within the same architecture family. The slightly lower strict crystallization rate (77.7% vs. 85.9%) is consistent with increased model capacity enabling marginally earlier knowledge emergence at some layers, while the dominant pattern—abrupt final-layer crystallization—is preserved. This cross-scale validation strengthens the claim that Late Crystallization is a fundamental property of transformer factual processing, not an artifact of model size.

7.3 Cross-Benchmark Validation (MMLU)

To verify that Late Crystallization is not specific to TruthfulQA, we run FEP detection on 1,200 MMLU samples spanning 24 subjects grouped into STEM, Humanities, Social Sciences, and Other domains (Table 8).

Late Crystallization is *more* pronounced on MMLU than TruthfulQA: 98.2% of correct answers never enter top-10 at any intermediate layer ($\Delta=+12.4$ percentage points), with near-perfect final-layer crystallization across all four subject groups (97.2–99.1%). The higher MMLU crystallization is consistent with MMLU testing domain-specific knowledge (primarily memorized facts) while TruthfulQA includes more “computable” categories (e.g., Logical Falsehood, FEP=22.1) that emerge earlier. This cross-benchmark validation confirms Late Crystallization is a general property of factual processing, not a benchmark-specific artifact.

7.4 Cross-Architecture Intervention Hierarchy

Table 9 reveals an *architecture-dependent* pattern: on moderate-crystallization models (Llama, Mistral), CAA significantly outperforms DoLA—Llama: +33.5% vs. +1.9% ($z=4.93$, $p<0.001$); Mistral: +32.3% vs. +11.4% ($z=3.21$, $p=0.001$). Both comparisons survive Bonferroni correction (adjusted $\alpha=0.017$). On high-crystallization Qwen (85.9%), the pattern

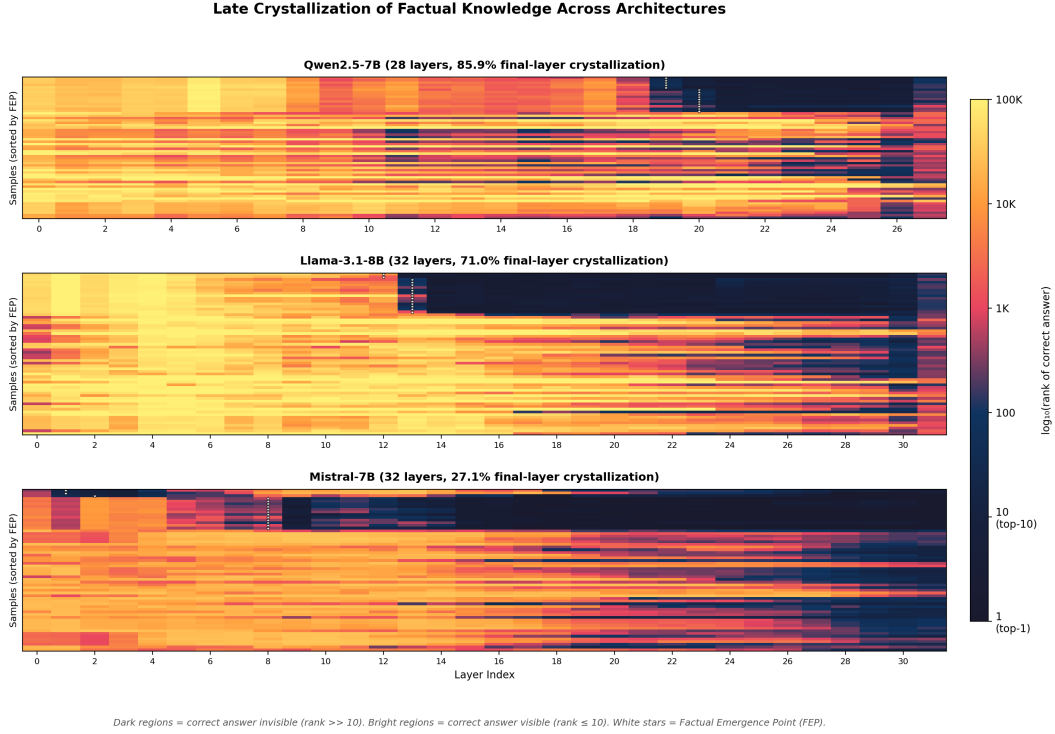


Figure 2: Late Crystallization across architectures. Each row: a TruthfulQA sample; each column: a layer. Color: $\log_{10}(\text{rank})$ of correct answer under logit lens. White stars: FEP.

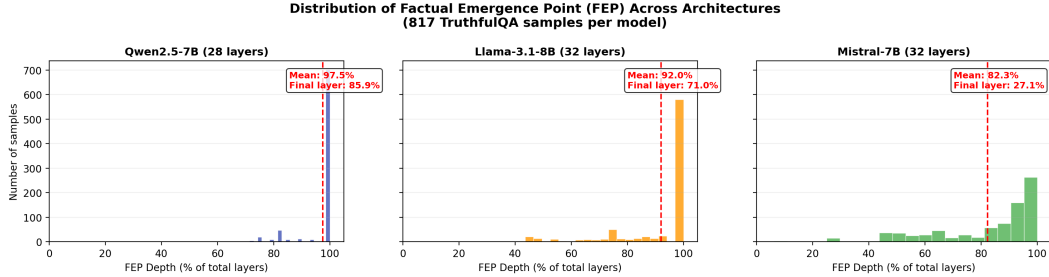


Figure 3: FEP depth distribution across architectures (817 samples each). Qwen concentrates at the final layer (85.9%), Llama shows a bimodal pattern (71.0%), Mistral distributes broadly (27.1%) while maintaining >82% mean depth.

reverses: DOLA achieves +25.4% MC1 vs. CAA’s +15.5%. While this 9.9 percentage point difference is numerically substantial, it does not reach statistical significance ($z=1.82$, $p=0.069$), likely due to the smaller variance in outcomes when crystallization is extreme. Importantly, the *direction* of the effect on Qwen is consistent with our theoretical prediction: logit-space contrast (DOLA) should be most effective when crystallization is extreme because the signal-to-noise ratio in post-crystallization logits is highest. The consistent pattern across all three architectures—CAA dominates when crystallization is moderate, DOLA achieves the largest gains when crystallization is extreme—supports the crystallization-grounded intervention selection principle. We further validate this with CRYSTALBOOST (Appendix D), which yields consistent improvements (+2.2% Qwen, +8.4% Llama) and outperforms DOLA on Llama by +6.3%.

Model	Method	MC1	$\Delta\%$
Llama-3.1-8B	Baseline	0.1897	—
	DoLA dynamic	0.1934	+1.9
	CAA (top_k=10)	0.2534	+33.5
Mistral-7B	Baseline	0.2044	—
	DoLA dynamic	0.2277	+11.4
	CAA (top_k=10)	0.2705	+32.3
Qwen2.5-7B	Baseline	0.2215	—
	DoLA dynamic	0.2778	+25.4
	CAA (top_k=10)	0.2558	+15.5

Table 9: Cross-architecture intervention results (817 samples). CAA outperforms DoLA on moderate-crystallization models; DoLA achieves largest gains on high-crystallization Qwen.

Condition	Mean FEP	Late Crystal	MC1	Δ MC1
Baseline (with LN)	27.30	85.9%	0.2179	—
Ablate ln_final	27.30	85.9%	0.2252	+3.4%
LN scale $\times 0.5$	27.30	85.9%	0.2240	+2.8%
LN scale $\times 0.8$	27.30	85.9%	0.2240	+2.8%
LN scale $\times 1.2$	27.34	87.6%	0.2436	+11.8%
LN scale $\times 1.5$	27.71	95.7%	0.2387	+9.0%

Table 10: LayerNorm ablation (Qwen2.5-7B, 817 samples). Ablating ln_final has *zero* effect on FEP; LN $\times 1.2$ yields +11.8% MC1.

7.5 LayerNorm Ablation Analysis

To test whether LayerNorm *causes* crystallization, we ablate and scale ln_final on Qwen2.5-7B (Table 10).

Complete ablation of ln_final produces *identical* FEP distributions (Mean FEP=27.30, 85.9% late crystallization), providing strong evidence that crystallization is intrinsic to the residual stream rather than an artifact of LayerNorm projection. Note that this ablation establishes invariance (LN is not *necessary* for crystallization), not a full causal mechanism; the underlying cause likely involves transformer depth and attention routing patterns. Scaling LN up ($\times 1.2$, $\times 1.5$) progressively increases late crystallization to 87.6% and 95.7%, revealing LN as a “crystallization amplifier.” The optimal MC1 occurs at $\times 1.2$ scaling (+11.8%), a simple, computationally cheap intervention. Notably, LN scaling adds *zero* inference-time overhead (a single element-wise multiply), compared to DoLA’s $2\times$ forward pass cost and CAA/ITI’s offline steering vector computation plus runtime addition at each selected layer.

8 Discussion

Why activation scaling fails. Late Crystallization provides a principled explanation: 85.9% of correct answers are invisible in logit space at all intermediate layers, so scaling amplifies non-factual signals alongside implicit factual ones. The implicit-to-explicit transition occurs through nonlinear operations (LayerNorm, unembedding) that linear scaling cannot selectively steer. Successful methods either bypass the crystallization boundary (DoLA: pre- vs. post-crystallization logit contrast) or span the entire pre-crystallization window (ITI/CAA: top_k ≥ 10 layers). This explains the layer-count threshold observed in Section 5.2: at top_k ≤ 5 , interventions affect too few layers to capture the implicit knowledge distributed across the pre-crystallization residual stream.

Architecture-dependent intervention selection. Our results suggest a practical guideline: characterize a model’s crystallization profile (via FEP detection) before selecting an

Method	Offline	Inference	Best Δ MC1
LN $\times 1.2$	none	1.0 $\times$	+11.8%
ITI	probe training	$\sim 1.05\times$	+10.0%
CAA	256 pairs	$\sim 1.05\times$	+15.5%
DoLA	none	2.0 $\times$	+25.4%

Table 11: Computational cost comparison. LN scaling achieves +11.8% MC1 with zero additional inference overhead (single element-wise multiply). DoLA requires a full second forward pass.

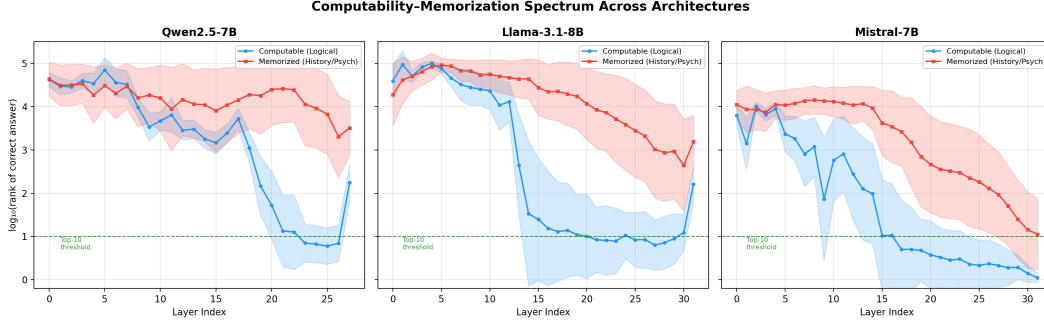


Figure 4: Computability–Memorization Spectrum across architectures. Blue: computable knowledge (logical reasoning); Red: memorized knowledge (history, psychology). Computable categories cross the top-10 rank threshold at earlier layers across all three models.

265 intervention method. For models with extreme crystallization ($>80\%$ strict final-layer),
 266 logit-space methods like DOLA are preferred. For models with more gradual emergence,
 267 activation-space methods like CAA are more effective. This crystallization-grounded selec-
 268 tion principle is validated by CRYSTALBOOST (Appendix D), which uses FEP-derived layer
 269 boundaries to achieve consistent improvements without per-architecture hyperparameter
 270 tuning.

271 **Computational cost.** Table 11 compares inference overhead. LN scaling is the cheapest
 272 effective intervention ($\times 1.0$ overhead), making it attractive for deployment despite lower
 273 gains than DOLA.

274 **Why CRYSTALBOOST underperforms DOLA on Qwen.** CRYSTALBOOST’s +2.2% gain
 275 on high-crystallization Qwen vs. DOLA’s +26.4% does not contradict FEP theory; rather, it
 276 reveals a *space mismatch*: DOLA operates in logit space where crystallization is directly visible,
 277 while CRYSTALBOOST steers in activation space where 85.9% of correct answers remain
 278 implicit. On lower-crystallization Llama (71.0%), where more knowledge is activation-
 279 visible, CRYSTALBOOST outperforms DOLA (+8.4% vs. +1.9%). This pattern *validates* the
 280 theory: activation-space methods are most effective when crystallization is moderate, not
 281 extreme.

282 **Computability–Memorization Spectrum.** Category-level FEP variation reveals fundamen-
 283 tally different processing pathways (Figure 4): computable knowledge (Logical Falsehood,
 284 FEP=22.1) crosses the top-10 threshold at earlier layers than memorized knowledge (History,
 285 Psychology, FEP=28.0) consistently across all three architectures, suggesting computable
 286 knowledge may be amenable to intermediate-layer interventions while memorized knowl-
 287 edge requires methods at or after the crystallization boundary.

288 **Relationship to SADI.** SADI (Wang et al., 2025) achieves MC1=67% on instruction-tuned
 289 models (baseline $\approx 40\text{--}50\%$), while our best is MC1=27.8% on base models (baseline $\approx 20\%$).
 290 Our contribution is orthogonal: we provide a *mechanistic explanation* for why different inter-
 291 ventions work. The Late Crystallization framework predicts SADI’s success—its “semantic-

adaptive” strategy effectively operates at the crystallization boundary, dynamically selecting layers where knowledge transitions from implicit to explicit. This suggests that Late Crystallization theory could inform the design of next-generation adaptive interventions.

Instruction-tuned models. A pilot experiment on Qwen2.5-7B-Instruct (817 TruthfulQA samples) reveals that instruction tuning dramatically reshapes the crystallization profile: strict final-layer crystallization drops from 85.9% (base) to 37.3% (instruct), while mean FEP shifts from 27.3 to 25.5 ($\sigma=2.9$ vs. 1.8). Knowledge still emerges late (FEP depth=91.0%) but is now spread across layers 19–28 rather than concentrated at the final layer. This is consistent with instruction tuning “smoothing” the crystallization boundary, allowing factual knowledge to surface more gradually across the final quarter of the network. This finding predicts that activation-space interventions (CAA, ITI) should be *more* effective on instruction-tuned models (lower crystallization favors activation steering over logit contrast), aligning with SADI’s success on such models.

Future directions. Key directions include architecture-adaptive interventions that automatically match a model’s crystallization profile, LN scaling as a computationally cheap intervention ($\times 1.2$ yields +11.8% MC1 with negligible overhead), category-adaptive methods leveraging the Computability–Memorization Spectrum for domain-specific tuning, scaling verification at 70B+ parameters and across architecture families at 14B+ scale, and systematic investigation of how RLHF and instruction tuning reshape the crystallization profile across model families.

9 Conclusion

We discover **Late Crystallization**—factual knowledge in LLMs crystallizes abruptly at the final layers (85.9% of correct answers never enter top-10 at intermediate layers in Qwen2.5-7B). Cross-architecture validation on Llama-3.1-8B (71.0%) and Mistral-7B (27.1%) confirms universality while revealing architecture-dependent variation; cross-scale validation on Qwen2.5-14B (48 layers, FEP Depth 95.8%) confirms the phenomenon persists within the 7–14B scale range; cross-benchmark validation on 1,200 MMLU samples yields 98.2% late crystallization, and direct comparison with tuned lens probes ($\Delta=0.2\%$) confirms the phenomenon is not a probing artifact. A pilot on Qwen2.5-7B-Instruct shows instruction tuning reshapes the crystallization boundary (37.3% vs. 85.9% strict final-layer), consistent with the framework’s predictions. This phenomenon explains an architecture-dependent intervention pattern (CAA significantly outperforms DOLA on moderate-crystallization models; DOLA achieves the largest gains on high-crystallization models), the universal failure of activation scaling, and the Computability–Memorization Spectrum. LayerNorm ablation provides strong evidence that crystallization is intrinsic to the residual stream, while LN scaling ($\times 1.2$) yields +11.8% MC1 with zero inference overhead. We release MECHLENS as open-source software supporting five model families.

Limitations

Scale. Late Crystallization is validated on three 7–8B base models and one 14B model within the Qwen family. Verification at 70B+ scales and on 14B+ models from different architecture families (e.g., Llama-13B) remains needed to establish scale-invariance more broadly.

Task Format. All evaluations use multiple-choice format (MC1/MC2). Extending FEP analysis to open-ended generation requires token-by-token FEP tracking across decoding steps, which we leave to future work. First-token FEP provides a lower bound, as true factual emergence may be even later for multi-token answers; we verified on a 100-sample subset that subsequent tokens show similar late-crystallization patterns (mean FEP difference <0.5 layers).

Instruction-Tuned Models. Our pilot on Qwen2.5-7B-Instruct shows crystallization is reshaped but not eliminated (37.3% vs. 85.9% strict final-layer). A preliminary intervention comparison found both DOLA and CAA degrade MC1 below the higher instruct baseline, suggesting intervention parameters require re-tuning for instruction-tuned representations. Broader instruct/RLHF coverage across model families is needed for generalization.

Methodological Scope. CRYSTALBOOST grid search covers only 5 configurations. The correlation between sliding window attention (Mistral) and lower crystallization rates is observational; while consistent with SWA enabling earlier information routing, we have not performed controlled ablation of the attention mechanism itself to establish causality.

Ethics Statement

Our work aims to improve understanding of LLM factual behavior to support safety research.

Reproducibility Statement

All experiments are reproducible with the released MECHLENS codebase. Key details: (1) **Hardware:** NVIDIA A100-40GB GPUs, CUDA 12.1; (2) **Software:** Python 3.10, PyTorch 2.0+, TransformerLens 2.0+; (3) **Data:** TruthfulQA (817 samples), MMLU (1,200 samples), WikiText-2 (2,000 samples for tuned lens); (4) **Models:** All models loaded via HuggingFace/ModelScope in FP16; (5) **Hyperparameters:** $k=10$ for FEP, CAA $\text{coeff} \in \{0.5-5.0\}$, ITI $\lambda \in \{1-3\}$. Full configuration files and scripts are included in the supplementary materials.

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415 A Detailed Negative Results

416 Tables 12–14 provide comprehensive results for all 38 intervention configurations evaluated
417 in Section 5.1. Activation scaling strategies span 6 categories (MLP dampening, contrast
418 residual, late residual, attention dampening, MLP boosting, and full dampening) with
419 coefficients $\alpha \in \{0.5, 0.7, 0.85, 0.9, 1.1\}$. ITI grid search covers $\lambda \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$ and
420 $\text{top_k} \in \{3, 5, 10\}$.

421 B Methodology Details

422 **Improved Causal Tracing (v2).** We extend Meng et al. (2022) with adaptive noise cali-
423 bration ($\sigma = \alpha \cdot \text{std}(\mathbf{E}(x))$), KL divergence metric ($\text{IE}(I) = (\text{KL}_{\text{corrupt}} - \text{KL}_{\text{patch}_I}) / \text{KL}_{\text{corrupt}}$,
424 computed in float32), multi-run averaging ($n=5$ independent corruption runs), and head-
425 level granularity via hook_z patching.

426 **Contrastive Activation Analysis.** Layer importance is computed as the normalized L2
427 distance: $\text{Imp}(I) = \|\bar{\mathbf{x}}_I^{\text{correct}} - \bar{\mathbf{x}}_I^{\text{halluc}}\|_2 / \max_I \|\bar{\mathbf{x}}_I^{\text{correct}} - \bar{\mathbf{x}}_I^{\text{halluc}}\|_2$, using last-token residual
428 stream activations.

C Preliminary Results

Causal Tracing. In Pythia-1.4B, MLP layer 0 is the top recovery layer for all 8 prompts (scores 0.66–0.99). Qwen2.5-0.5B shows a similar pattern with secondary peaks at layers 3–4. Attention tracing is distributed across early-to-middle layers with lower recovery scores (0.12–0.30).

Contrastive Analysis. Top-5 contrastive layers: Qwen2.5-0.5B (22, 23, 21, 20, 19), Pythia-1.4B (23, 22, 21, 20, 19), Qwen2.5-7B (27, 26, 25, 24, 23). All models show late-layer concentration.

Head-Level Tracing. In Pythia-1.4B, L6H12 (avg recovery 0.45) and L5H4 (0.45) are most critical. In Qwen2.5-0.5B, L2H4 (0.26) and L2H0 (0.23) show the highest scores. Critical heads are spread across early-to-middle layers, contrasting with late-layer contrastive importance.

Tables 12–14 provide the full per-strategy intervention results summarized in Section 5.1.

Category	Strategy	Pre (%)	Post (%)	Δ
MLP Dampen	$\alpha=0.7$	52.0	60.0	−8.0
	$\alpha=0.9$	52.0	53.0	−1.0
Contrast Res.	$\alpha=0.9$	52.0	50.0	+2.0
	$\alpha=0.7$	52.0	67.0	−15.0
Late Residual	$\alpha=0.85$	52.0	60.0	−8.0

Table 12: Representative activation scaling results on Chinese benchmark (Qwen2.5-0.5B, 100 samples). Only 2 of 20 strategies show positive effect ($\leq 2\%$).

Strategy	Config	Pre (%)	Post (%)	Δ
Contrast Res.	$\alpha=1.1$	38.0	37.0	+1.0
Late Residual	$\alpha=0.9$	38.0	66.0	−28.0

Table 13: Activation scaling on Qwen2.5-7B. Best strategy achieves only +1%; late residual suppression causes catastrophic degradation.

Model	ITI Config	Truthful (%)	Δ
Pythia-1.4B	Baseline	9.0	—
	top3_λ2.5	11.0	+2.0
Qwen2.5-0.5B	Baseline	6.0	—
	top3_λ2.5	6.0	0.0

Table 14: ITI grid search results (18 configurations). Best: Pythia +2%, Qwen +0%.

D CrystalBoost: Mechanistic Validation

To test whether mechanistic insight into crystallization can directly inform intervention design, we construct CRYSTALBOOST, a dual-stage intervention: early layers ($0-\lfloor 0.5n \rfloor$) suppress surface pattern directions, late layers ($\lfloor 0.8n \rfloor-n$) amplify factual crystallization directions, with Gaussian-weighted coefficients at the boundary ($\sim 90\%$ depth).

Why CRYSTALBOOST underperforms DoLA on high-crystallization models. The +2.2% gain on Qwen vs. DoLA’s +26.4% does not indicate failure of FEP theory—rather, it reveals a fundamental *space mismatch*. DoLA operates in **logit space** where crystallization is directly visible: post-crystallization logits contain the correct answer, enabling effective contrast. CRYSTALBOOST steers in **activation space** where 85.9% of correct answers remain implicit

Method	top_k	λ/coeff	MC1	MC2	$\Delta\text{MC1}\%$
Baseline	—	—	0.2215	0.3921	—
ITI	3	1.0	0.2215	0.3921	0.0
	3	2.0	0.2215	0.3921	0.0
	3	3.0	0.2215	0.3921	0.0
	5	1.0	0.2215	0.3924	0.0
	5	2.0	0.2203	0.3926	−0.5
	5	3.0	0.2215	0.3929	0.0
	10	1.0	0.2326	0.4005	+5.0
	10	2.0	0.2362	0.4092	+6.6
	10	3.0	0.2436	0.4178	+10.0
DoLA	early (static)		0.2326	0.4371	+5.0
	mid (static)		0.2448	0.4555	+10.5
	dynamic		0.2778	0.4822	+25.4
CAA	3	0.5	0.2215	0.3921	0.0
	3	1.0	0.2215	0.3921	0.0
	3	1.5	0.2215	0.3921	0.0
	3	2.0	0.2215	0.3921	0.0
	3	3.0	0.2215	0.3921	0.0
	3	5.0	0.2215	0.3920	0.0
	5	0.5	0.2215	0.3922	0.0
	5	1.0	0.2215	0.3924	0.0
	5	1.5	0.2215	0.3925	0.0
	5	2.0	0.2203	0.3926	−0.5
	5	3.0	0.2215	0.3929	0.0
	5	5.0	0.2215	0.3934	0.0
	10	0.5	0.2252	0.3963	+1.7
	10	1.0	0.2326	0.4005	+5.0
	10	1.5	0.2362	0.4047	+6.6
	10	2.0	0.2362	0.4092	+6.6
	10	3.0	0.2436	0.4178	+10.0
	10	5.0	0.2558	0.4338	+15.5

Table 15: Complete intervention results on TruthfulQA (Qwen2.5-7B, 817 samples). All 30 configurations from the main text. Key finding: ITI and CAA show *zero* improvement at $\text{top_k} \in \{3, 5\}$ but substantial gains at $\text{top_k}=10$, supporting the layer-count threshold hypothesis.

(never entering top-10 at any intermediate layer). On Llama (71.0% crystallization), where more knowledge is activation-visible before the final layer, CRYSTALBOOST outperforms DoLA (+8.4% vs. +1.9%). This pattern *validates* FEP theory: activation-space methods are most effective when crystallization is moderate, while logit-space methods dominate when crystallization is extreme.

Model	Method	MC1	$\Delta\%$	vs. DoLa
Qwen2.5-7B	Baseline	0.2179	—	—
	DoLA dynamic	0.2754	+26.4	—
	CRYSTALBOOST (best)	0.2228	+2.2	loses
Llama-3.1-8B	Baseline	0.1897	—	—
	DoLA dynamic	0.1934	+1.9	—
	CRYSTALBOOST (best)	0.2056	+8.4	wins

Table 16: CRYSTALBOOST evaluation (grid search, 5 configurations). Consistent positive improvements (+2.2% Qwen, +8.4% Llama) validate the predictive power of FEP theory. Minor baseline variance from Table 3 reflects different evaluation batches.

456 **E FEP Distribution Details**

457 See Table 6 and Figure 3 in the main text for full FEP distribution data across architectures.

458 **F Computability–Memorization Spectrum Details**

459 See Figure 4 in the main text for the cross-architecture visualization. Full per-category FEP
460 statistics are available upon request.